101

HUMBLE MAX

ZZB 111: ANIMAL TAXONOMY LECTURE NOTES

CLASSIFICATION

Objectives

After completing this section, you should know:

- *how to write the scientific name of animal correctly*
- *know that animals belong to the Animal kingdom and that this is divided into phyla, classes, orders, families, genus and species.*
- *know the definition of a species*
- *know the phylum and class of the more common animals dealt with in this course*

Classification of Living Organisms

There are over 2,000,000 known different organisms in the world today and probably over 1,000,000 are left to be discovered and named. All these living things have different shape, size color ... it would be extremely difficult to study the ... of living things without arranging them into groups. Classification (taxonomy) helps to give a universal scientific name to every species so that students of different languages know the scientific name of that species. Common names vary from language to language ... scientific names are internationally accepted. Biologists, in arranging living things into groups have developed system of using similarities to unit species into groups and using differences to separate groups. The science of arranging living things into groups based on structural, genetic or cellular organization is called classification or taxonomy. Example of structural similarities and differences exist in fish, frog, lizard, bird and man. They are grouped together as vertebrates because they all posses backbone (vertebral column). On the other hand, they grouped differently because frog has a wet and scale-less skin, lizard has a dry skin ... bird has beak and feathers while man has hair and mammary glands. The relationship between organisms can also account for grouping. If two animals have the same number of similar chromosomes, then the two animals are closely related and can be grouped together. For example, Gorilla and Chimpanzee have 48 chromosomes each hence they are grouped into the family Pongidea. The same reason goes for cellular organization for grouping. For example, all animals with one cell are grouped as protozoa (unicellular organisms) and those with more than one cell are grouped as metazoa (multicellular organisms). Initially this classification was based only on the appearance of the organism. However, the development of new techniques has advanced our scientific knowledge. The light microscope and later the electron microscope have enabled us to view the smallest structures, and now techniques for comparing DNA have begun to clarify still further the relationships between organisms. In the light of the advances in knowledge the classification has undergone numerous revisions over time. At present most biologists divide the living world into 5 kingdoms, namely:

i. Bacteria ii. Protists iii. Fungi iv. Plants v. Animal

Classification is the process used by scientists to make sense of the over 2 million or so different kinds of living organisms on the planet. It does this by describing, naming and placing them in different groups. As zoologist, you are mainly concerned with the Animal Kingdom but don't forget that animals rely on the Plant Kingdom for food to survive. Also many diseases that animals are affected by, are members of the other Kingdoms -- fungi, bacteria and single celled animals.

Taxonomic Grouping

Living things are first separated into large groups called Kingdoms. The division of kingdoms to smaller groups is called **Phyla** (singular Phylum). Phyla are divided into smaller groups called classes. Each class is further divided into orders. Orders are further divided into families. Families are again divided into genera (singular genus). A genus is broken into species.

A kingdom is group of related phyla
A phylum is a group of related classes
A class is a group of related order
An order is group of related family
A family is a group of genera
A genus is a group of a closely related species
A species is a population of related organisms that can interbreed within themselves to produce fertile offspring.

Naming Animals

Of course we know what a cat, a dog and a whale are but, in some situations using the common names for animals can be confusing. Problems arise because people in different countries, and even sometimes in the same country, have different common names for the same animals. For example, a cat can be a chat, a Katze, gato, katt, or a moggie, depending on which language you use. To add to the confusion sometimes the same name is used for different animals. For example, the name 'gopher' is used for ground squirrels, rodents (pocket gophers), for moles and in the south-eastern United States for a turtle. This is the reason why all animals have been given an official scientific or binomial name. Unfortunately, these names are always in Latin. For example:

- Common rat: *Rattus rattus*
- Human: *Homo sapiens*
- Domestic cat: *Felis domesticus*
- Domestic dog: *Canis familiaris*
- Lion: *Panthera leo*
- Housefly: *Musca domestica*

The first time you refer to an organism, you should write the whole name in full. If you need to keep referring to the same organism, you can then abbreviate the genus name to just the initial. Thus "*Canis familiaris*" becomes "*C. familiaris*" the second and subsequent times you refer to it.

As you can see from the above there are certain rules about writing scientific names:

- They always have 2 parts.
- The first part is the genus name and is always written with a capital first letter.
- The second name is the species name and is always written in lower case.
- The name is always **underlined** or printed in italics.

The Animal Kingdom

So what are animals? If we were suddenly confronted with an animal we had never seen in our lives before, how would we know it was not a plant or even a fungus? We all intuitively know part of the answer to this.

Animals:

- eat organic materials (plants or other animals)
- move to find food
- take the food into their bodies and then digest it
- and most reproduce by fertilizing eggs by sperm

If you were tempted to add that animals are furry, run around on four legs and give birth to young that they feed on milk you were thinking only of mammals and forgetting temporarily that frogs, snakes and crocodiles, birds as well as fish, are also animals.

These are all members of the group called the **vertebrates** (or animals with a backbone) and mammals make up only about 8% of this group. The diagram on the next page shows the percentage of the different kinds of vertebrates.

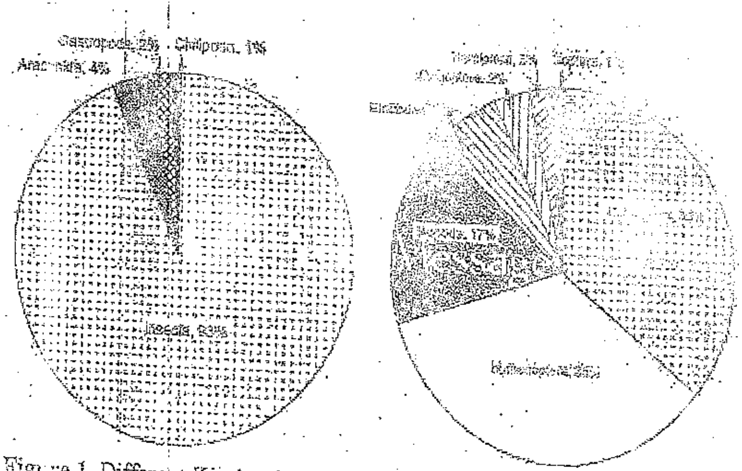


Figure 1. Different Kinds of Invertebrates

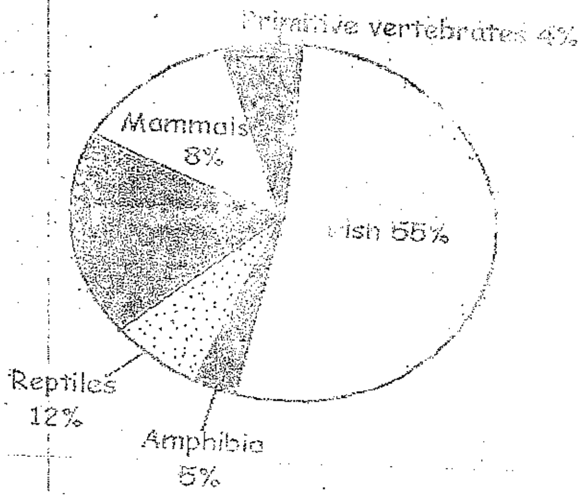
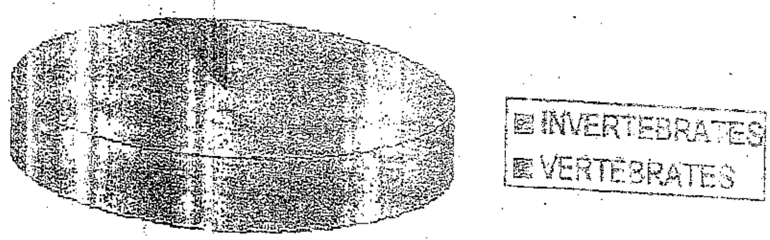


Figure 2 Different Kinds of Vertebrate

However, the term animal includes much more than just the Vertebrates. In fact, this group makes up only a very small portion of all animals. Take a look at the diagram below, which shows the size of the different groups of animals in the Animal Kingdom as proportions of the total number of different animal species. Notice the small size of the segment representing vertebrates! All the other animals in the Animal Kingdom are animals with no backbone, or invertebrates. This includes the worms, sea anemones, starfish, snails, crabs, spiders and insects. As more than 90% of the invertebrates are insects, no wonder people worry that insects may take over the world one day!



Figur 3

Classification of Animals

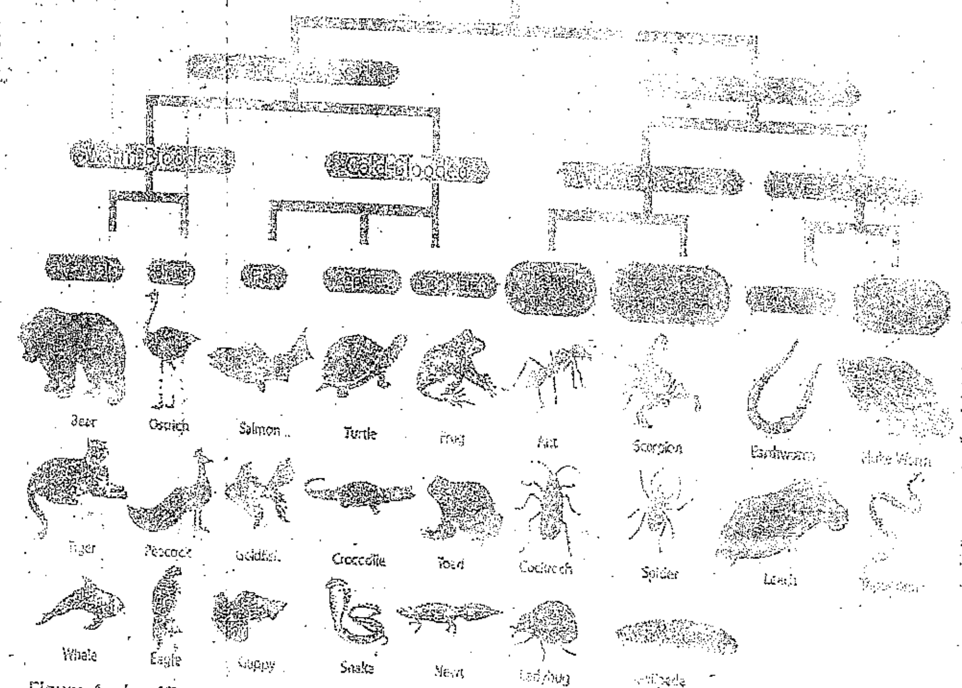


Figure 4. classification of animals

The Classification of Vertebrates

As we have seen above the Vertebrates are divided into 5 groups or classes namely:

- Fish (pisces)
- Amphibian (frogs and toads)
- Reptiles (snakes and crocodiles)
- Birds (Aves)
- Mammals

These classes are all based on similarities. For instance, all mammals have a similar skeleton, hair on their bodies, are warm bodied and suckle their young.

The class Mammalia (the mammals) contains 3 **subclasses**:

- Duck billed platypus and the spiny anteater
- **Marsupials** (animals like the kangaroo with pouches)
- **True mammals** (with a placenta)

Within the subclass containing the true mammals, there are groupings called **orders** that contain mammals that are more closely similar or related, than others. Examples of six mammalian orders are given below:

- Rodents (Rodentia) (rats and mice)
- Carnivores (Carnivora) (cats, dogs, bears and seals)
- Even-toed grazers (Artiodactyla) (pigs, sheep, cattle, antelopes)
- Odd-toed grazers (Perissodactyla) (horses, donkeys, zebras)
- Marine mammals (Cetacea) (whales, sea cows)
- Primates (monkeys, apes, humans)

Within each order there are various **families**. For example, within the carnivore mammals are the families:

- Canidae (dog-like carnivores)
- Felidae (cat-like carnivores)

Even at this point it is possible to find groupings that are more closely related than others. These groups are called **genera** (singular genus). For instance, within the cat family Felidae is the genus Felis containing the cats, as well as genera containing panthers, lynxes, and sabre toothed tigers.

The final groups within the system are the species. A species is a group of animals that can mate successfully and produce fertile offspring. This means that all domestic cats belong to the species *Felis domesticus*, because all breeds of cat whether Siamese, Manx or ordinary House cat can cross breed. However, domestic cats cannot mate successfully with lions, tigers or jaguars, so these are placed in separate species, e.g. *Felis leo, Felis tigris and Felis* [illegible].

Even within the same species, there can be animals with quite wide variations in appearance that still breed successfully. We call these different breeds, races or **varieties**. For example, there are many different breeds of dogs from Dalmatian to Chihuahua and of cats, from Siamese to Manx and domestic short-hairs, but all can cross breed. Often these breeds have been produced by selective breeding but varieties can arise in the wild when groups of animals are separated by a mountain range or sea and have developed different characteristics over long periods of time.

Invertebrates

An invertebrate refers to any of the animals lacking a vertebral column.

Examples of Invertebrates

Some of the major groups of invertebrates are as follows:

- Porifera (sponges) – characterized by their lack of true tissues and organs.
- Cnidaria (sea anemones, jellyfish, corals, and box jellies) – exhibit two morphological body plans, i.e. polyp or medusa. They have specialized cells called cnidocytes that contain nematocysts used for immobilizing prey. Most of them are marine species.
- Platyhelminthes (flatworms) – invertebrates that are acoelomates, and lacking complex circulatory and respiratory systems. Some groups reproduce asexually.
- Nematoda (roundworms) – invertebrates with cylindrical bodies. Many of them are parasites.
- Annelida (segmented worms) – invertebrates that have a segmented body plan
- Arthropoda (insects, spiders, crabs, etc.) – invertebrates with a hard external exoskeleton that is periodically shed and replaced. They have jointed legs. They are the largest animal phylum.
- Mollusca (cuttlefish, snails, mussels, etc.) – invertebrates with a muscular foot that they use for anchorage. They also have a mantle.
- Echinodermata (starfish, sea cucumbers, etc.) – invertebrates with "spiny" skin

General Characteristics of invertebrates

- Invertebrates do not have backbones. This is the major characteristic that separates invertebrates from vertebrates.
- Invertebrates have less complex nervous systems.
- Invertebrates are cold-blooded animals.
- Invertebrates have distinct body parts.

Phylum: Protozoa
Characteristic features
- They belong to the group of organisms called Protista
- They are microscopic organisms
- They have eukaryotic cells i.e. cells with membrane
- They are one-celled or colony of one-celled animals
- They are either aquatic or endoparasitic in habitat
- They are free-living or parasitic organisms
- They reproduce asexually by binary fission

Examples of protozoa include: Amoeba, Paramecium and Volvox (aquatic), Trypanosome and plasmodium (endoparasites)

Phylum: Coelenterata

Characteristic features

HUMBEE MAX

- They are multicellular organisms
- They have two distinct body layers (diploblastic), ectoderm (outer layer) and endoderm (inner layer), separated by a featureless middle layer called mesoglea.
- Their bodies possess radial symmetry
- They have soft-like jelly bodies
- They are mainly aquatic organisms .
- They have only one body cavity called enteron
- There is only one opening called mouth to the outside world. No anus
- Tentacles surround the mouth
- Possession of stinging cells called nematocysts
- They have specialized cells such as glandular, nerve, sensory and absorptive cells
- Asexual reproduction by budding

Examples are Hydra, Obelia, Sea anemone, Jelly fish, Coral and Physalia

Phylum: Platyhelminthes (flat worms)

Characteristic features

- They are multicellular flat worms
- They are bilaterally symmetrical
- They do not have body cavity (acoelomate)
- They have soft flat body
- They possess alimentary canal with highly branched diverticula. No anus
- They are mainly parasites in man and other animals e.g. liver flukes and tapeworms while others are free living e.g. planaria
- Most flat worm hermaphrodites and reproduce sexually

Examples include: tape worm, liver flukes (fasciola), and blood fluke (schistosoma) and planaria

Phylum: Nematoda (round worms)

Characteristic features

- They have soft, cylindrical and slender bodies pointed at both ends
- They body is not segmented
- They have alimentary canal with both mouth and anus
- They are bilaterally symmetrical
- They have false body cavity (pseudocoelomates)
- Some are hermaphrodites while others reproduce sexually
- They are mostly parasitic but few are free living

Examples are Ascaris (round worm), hookworms, threadworms, guinea worms (parasites) and Vinegar eel.(free living)

Phylum: Annelida

Characteristic features

- They possess metamerically segmented body
- They body is cylindrical in shape
- They have a true body cavity (coelom) with distinct head
- They have alimentary canal with mouth and anus
- They appendages (setae) are not jointed
- Some are aquatic while others are terrestrial i.e. they live in the soil
- Annelids reproduce sexually and many are hermaphrodites
- They bodies are made up of three thick layers

Examples are the earthworms, leeches, and tube worms

Phylum: Mollusca

Characteristic features

- They possess soft and unsegmented bodies
- Tentacles are present in most members
- They possess muscular foot adapted for crawling or burrowing
- They body is covered by a soft tissue called mantle

Some have calcareous shells e.g. snails while other have no shells e.g. octopus and [illegible]

- Some are aquatic while other are t restrial
- They eye and tentacles are used fo sensitivity

Examples are squid, mussel, periwinkles a d snails

Phylum: Echinodermata

Characteristic features:

- They possess tough spiny and cal eous exoskeleton
- They are radially symmetrical in a ult but bilaterally symmetrical in larvae
- They head is not usually distinct a d the body is not segmented
- Most of them are star shape
- They are all marine
- Has tube feet which is used for mo ement
- They are triploblastic animals i.e. ey have three body layers

Examples are star fish, sea urchins, sea cu umbers, sea lily and bristle star

Phylum: Arthropoda

The Arthropod is the largest phylum in the nimal kingdom. it is divided into the following classes: (i) crustacean, (ii) Insecta, (iii). Arachnid (iv). Myriapoda

Characteristic features

- They have segmented bodies
- They bodies is covered with hard, gid exoskeleton made of chitin
- They jointed appendages or joi ed legs used for feeding, movement, reproductive activities or as sensory organs
- Often, the bodies are divided into t ree regions – head, thorax and abdomen. In some, the head and thorax are fused together to form cephalothorax as in crustaceans and arachnids
- They bodies are bilaterally symme ical
- They exhibit molting or ecalysis i.e shed their exoskeleton at interval to permit growth
- They are triploblastic i.e. they have three body layers
- They have various means of respi io i e.g. gills, trachea, lung book or body surface
- Some are aquatic while others are rrestrial

Examples are insects (mosquitoes and coc roach), crustaceans (e.g. crab and lobster), diplopods (e.g. millipede), chilopods (e.g. centipede) nd arachnids (e.g. spider and tick).

Vertebrates

The vertebrate is sub-phylum of the phy n chordata. The vertebrates are characterized by the presence of a backbone or vertebral colu n. They are divided into five classes namely: Pisces (fishes), Amphibia (amphibians), Reptila (ptiles), Aves (birds) and Mammalia (mammals)

These classes of vertebrates differ in so e ways although they have some characteristics in common.

General Characteristics of Vertebrates

- They possess an internal jointed sk eton made up of cartilage and/or bones
- They have bilateral symmetrical bo ies
- They body is divided into head, tru k and tail
- They have two pairs of limbs. The p ctoral limbs form fore-limbs or wings while the pelvic limbs form the hind limbs or legs
- They have a closed blood system w ich comprises the blood vessels and the heart
- They are triploblastic animals
- They possess skin which may be na ed or have a covering of scales, feathers or hairs

Class: Pisces (fishes)

Characteristic features

- They are aquatic animals; i.e. they ar be found in marine or fresh water, ponds lakes, rivers etc.
- They possess gills for respiration th ughout life
- The skin is covered by scales but fe are without ...

- They have fins which are used for movement in water
- They are cold blooded (Poikilothermic) animals i.e. their body temperature varies with that of their environment
- They have lateral line system. This is used for detection of vibration and pressure in water
- They have swim bladder which enables them to maintain buoyancy in water
- Reproduction is sexual and they have external types of fertilization
- They are oviparous animals i.e. they lay eggs which develop to adult stage outside the body of the adult female fish
- They have two-chambered heart

Pisces can sub-divided into two groups based on the nature of their skeletal system – bony fish and cartilaginous fish

Bony fish

These are fish with bony skeleton e.g. Tilapia, Carp, Salmon, Mackerel, Herring etc.

Cartilaginous fish

These fish whose bones are made of cartilages e.g. Dog fish, Shark, Skates, Stingrays etc.

Class: Amphibia

Characteristic features

- Possess moist, soft and scaleless skin
- They are cold blooded (Poikilothermic)
- Adults possess paired limbs with webbed toes
- The larval stage is always aquatic while the adult is either aquatic or terrestrial. Tadpole larva is vegetarian while adult is usually carnivorous.
- The heart is two chambered in larval stage and three chambered in adult
- They carryout gaseous exchange (respiration) by gills, lungs, skin and mouth
- The have sticky tongue which can be protruded or retracted quickly
- They have poisonous glands on their skin which are used for defence

Examples are toads, frogs, salamander, newts etc.

Class: Reptilia

Characteristic features

- They possess dry and scaly skin
- Most members possess four which ends in claws while others (snake) have no limbs
- Dentition is homodont (the same types of teeth)
- They are cold blooded (poikilothermic)
- Breathing is by lungs
- Fertilization is internal with laying of soft and leathery shelled eggs. Most members are oviparous while a few other retain their eggs in the body until hatching (ovoviviparous)
- They are mostly terrestrials but few such as the turtles and crocodile are aquatic
- The heart is three chambered except crocodile with four chambered heart

Examples are lizard, wall geckos, tortoise, snakes, crocodile, turtles, chameleon etc.

Class: Aves

Characteristic features

- They are homoeothermic or warm-blooded animals i.e. they have constant body temperature and very active animals
- Their entire body is covered with feathers except the legs and the feet, which are covered with scales
- Their fore limbs are modified into horny toothless beak or bill
- The bones are porous, light and in cases filled with air
- They bodies is supported by two hind limbs (bipedal stance)
- Breathing is by lungs they lay hard shelled eggs (oviparous)
- Fertilization is internal
- They are mostly terrestrial
- The heart is four chambered
- They show parental care

Examples are domestic fowl, duck, eagle, ostrich, weaver bird, woodpeckers' hawks' sparrow, pigeon etc

Class: Mammalia

Characteristic features

- They are homoiothermic or warm-blooded animals hence, they are very active
- Their bodies are covered with hair
- They possess external ear called pinnae
- They possess mammary glands. The mother's mammary glands secrete milk used in nourishing the young after birth
- They are viviparous i.e. the young are born at an advanced stage of development, however, the monotremes lay eggs which hatch into young, nourished by milk secreted by mammary glands of the mother
- They have diaphragm (breathing muscle) separating the thoracic cavity from the abdominal cavity
- Their heart is four chambered
- Most members possess forelimbs for locomotion
- Reproduction is sexual and fertilization is internal
- They have well developed brain
- They show parental care to the young ones

Examples are duck-bill platypus, rabbit, whale, bats, monkey, goat, lion, elephant, giraffe, gorilla and man.